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BRAC UNIVERSITY

CSE360: Computer Interfacing

Lab Project Report

Title: *LPG Cylinder Monitoring and Gas Leakage Detection System*
by

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Abstract:

This is a Microcontroller (Arduino Uno R3) Based LPG Cylinder Monitoring and Gas Leakage Detection System. It aims to prevent gas leaks and monitor gas consumption in real-time. The embedded setup is based on an Arduino Uno R3 that receives data from an MQ-2 gas sensor and a load cell with an HX711 amplifier. It provides real-time information on a LCD. The prototype is designed for emergencies. It automatically turns on an alarm and a fan for ventilation if it sounds an alarm and activates an exhaust fan. Also, if the weight of the gas cylinder is reduced, it displays low gas. The system has been tested and shown to detect leaks and weight changes of the cylinder. Apart from ensuring safety from fires, the system also helps in efficient resource utilisation and the wastage of partially filled cylinders.

Keywords

Arduino, Gas Leak Detection, LPG, Microcontroller, Safety Automation

1. Introduction

Problem and Motivation: In our current modern system energy and resources are often wasted because devices stay on most of the time or are controlled manually and due to this inefficiency it not only increases the costs but also harms the environment and for that we need a way for the system to sense their surroundings and make smart decisions automatically.

Project Objectives: The main goal of this project is to design such a system that allows a computer to connect to physical sensors and devices so that they can communicate with each other.

- Build a stable connection between hardware and software.
- Create a system that reacts immediately to environmental changes.
- Ensure the design is energy efficient, affordable and expandable for future proof.

Scope and Relevance: This project falls under Computer Interfacing as we can see it focuses on how data moves from the physical sensors into a digital processor and by applying such measures we can create a smarter system for everything and it can be from home automation to eco friendly factories**.**

2. Related Work/ Inspiration:

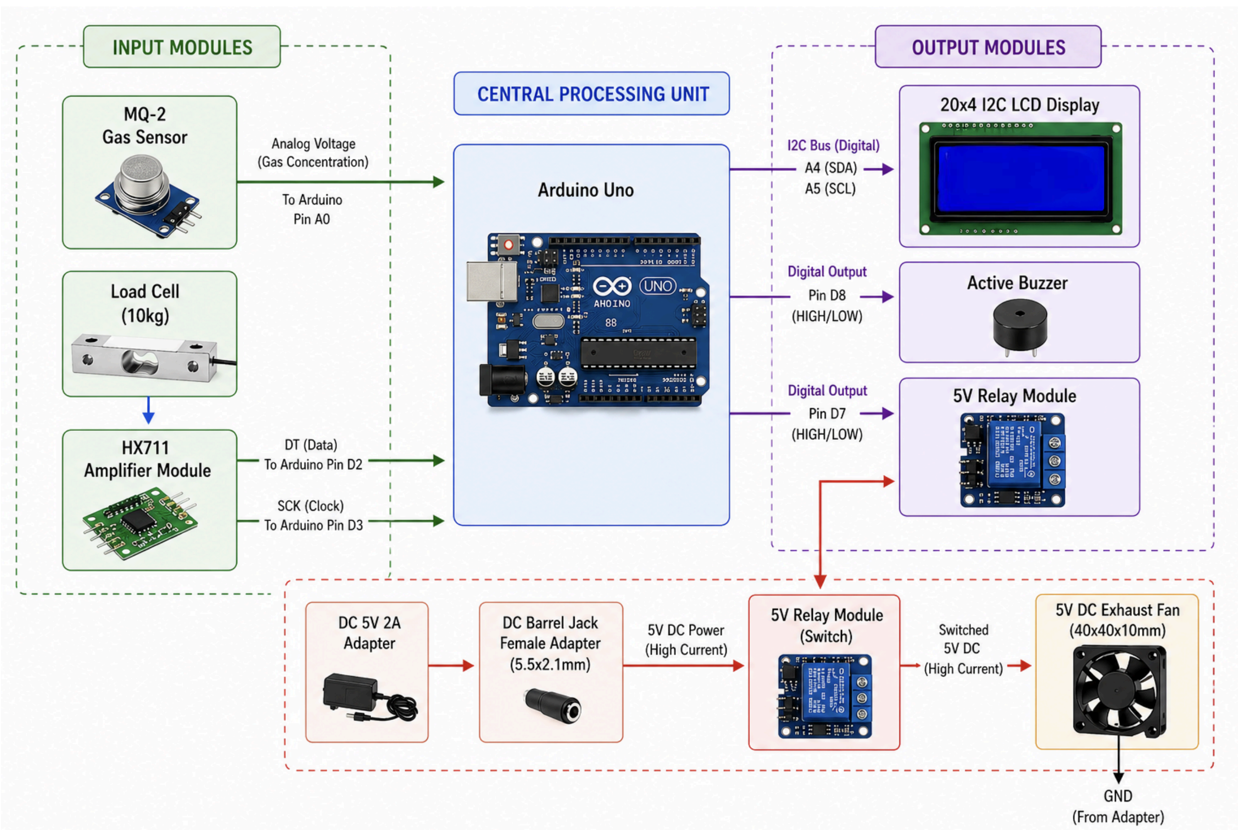
Existing Systems and Related Work: Oo et al. (2021) proposed an IoT-based LPG monitoring and leakage detection system using an Arduino platform; they used a load cell and an HX711 amplifier to determine the weight of the cylinder and a gas sensor to detect leaks. In addition, a fan was automatically turned on when leakage occurred to vent the gas. It also uses IoT for cloud monitoring and alarms to allow remote monitoring and control.Although the system by Oo et al. (2021) opts IoT for cloud based monitoring and remote alarms the proposed system is an affordable and system that does not require any internet and this makes it more accessible for families who can't afford it and who may not have proper internet access. Moreover, the design offers real time on site monitoring and immediate action through an LCD screen, alarm, and an exhaust fan. But the system is still open to further staff use of IoT components like ESP8266 or ESP32 for remote monitoring and alerts in the future.

Problem Gap and Contribution: While various solutions for LPG monitoring and gas leakage detection have been out there, most of them depend on complicated systems. This presents challenges in areas where there is no internet/costing is high or lack of access.Furthermore, these systems only include gas leakage detection that's it and without real time LPG weight monitoring system. Our system addresses these issues into consideration and we offer an affordable system that integrates both gas leakage detection and weight monitoring of the cylinders. In addition, it features a local alarm with a buzzer and automatic exhaust fan activation without any network connection.To sum it up, although the system doesn't represent new ideas, it provides an affordable solution by focusing on being affordable and real time systems while opting for future upgrades such as IoT.

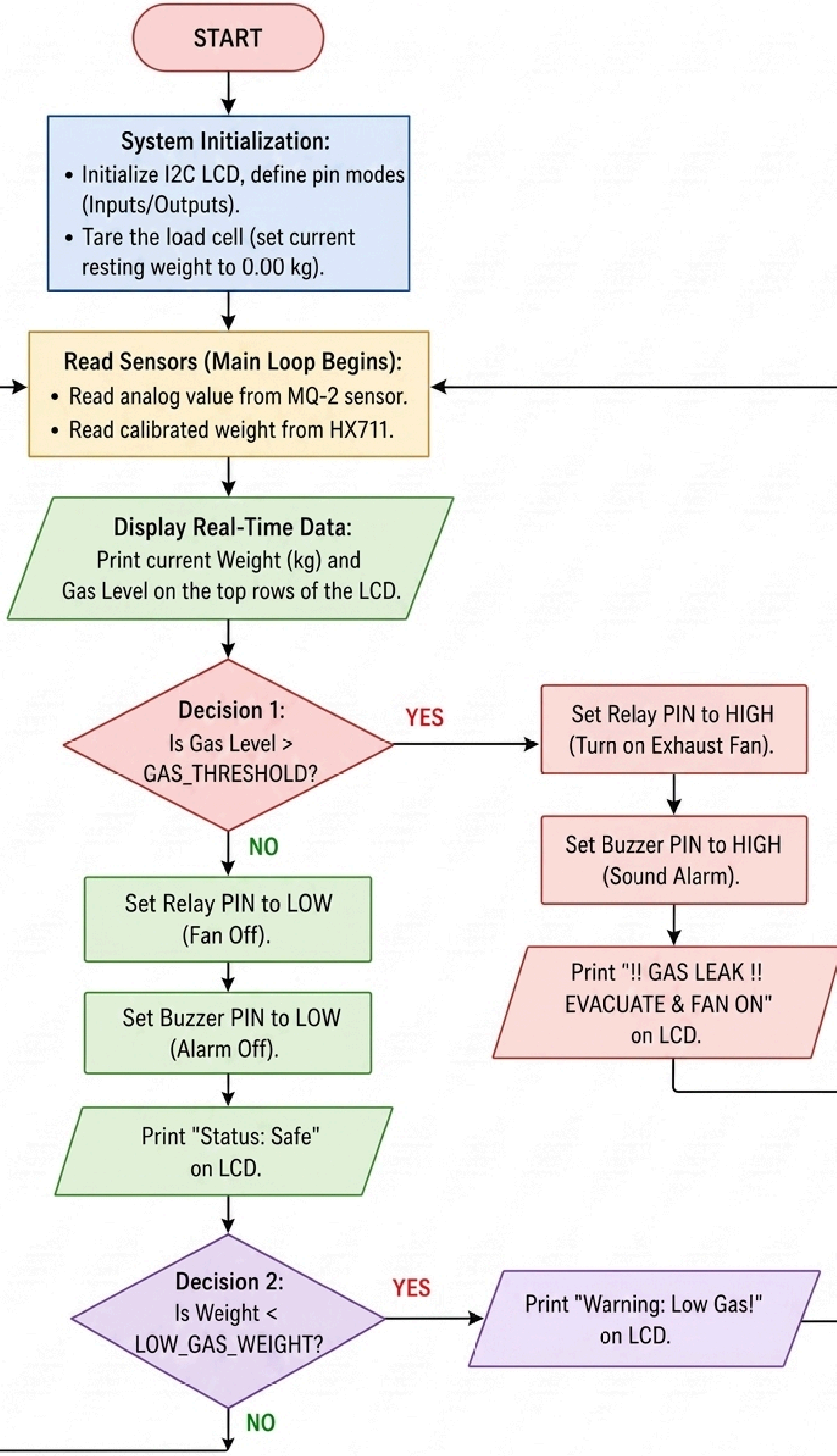
3. Technical Approach

System Architecture: Arduino Uno is like the brain that handles everything. The MQ-2 sensor keeps checking the air for gas all the time. A load cell, which is connected to an HX711 amplifier, measures how heavy the gas cylinder is. The LCD screen shows stats. Additionally, there is a loud buzzer and a 5V relay that, in an emergency, turns on a DC exhaust fan.

This design aims to maintain environmental safety. If there is a lot of gas, the mechanism activates the fan and buzzer. the fans and buzzer to protect us. If it's all good, it monitors the gas level in the gas tank. If the gas is low, it displays a message on the LCD screen.



*All components (MQ-2, HX711, LCD, Relay) have common 5V and GND from Arduino/Breadboard


START
System Initialization:
• Initialize I2C LCD, define pin modes (Inputs/Outputs).
• Tare the load cell (set current resting weight to 0.00 kg).
Read Sensors (Main Loop Begins):
• Read analog value from MQ-2 sensor.
• Read calibrated weight from HX711.
Display Real-Time Data:
Print current Weight (kg) and Gas Level on the top rows of the LCD.
Decision 1:
Is Gas Level > GAS_THRESHOLD?
YES
Set Relay PIN to HIGH (Turn on Exhaust Fan).
Set Buzzer PIN to HIGH (Sound Alarm).
Print "!! GAS LEAK !! EVACUATE & FAN ON" on LCD.
NO
Set Relay PIN to LOW (Fan Off).
Set Buzzer PIN to LOW (Alarm Off).
Print "Status: Safe" on LCD.
Decision 2:
Is Weight < LOW_GAS_WEIGHT?
YES
Print "Warning: Low Gas!" on LCD.
NO

Components Used:

Hardware Components:

- Arduino Uno R3 – A microcontroller board used to run the project.
- Load Cell Weight Sensor (10kg TAL220) – Measures weight.
- HX711 Load Cell Amplifier Module – Amplifies and converts load cell signals.
- MQ-2 Gas Sensor – Detects gases like LPG.
- LCD Display (I2C 20x4**)** – Displays data.
- Buzzer (Active) – Used for alerts.
- 5V Cooling Fan (40 x 40 x 10mm) – Provides airflow to reduce gas concentration
- 5V Relay Module for Arduino – Acts as a switch to control high-power
- Adapter DC 5V 2A 2.1mm Barrel Jack Adapter – Supplies stable 5V power
- DC Barrel Power Jack 5.5x2.1mm Female Adapter – Connects the power adapter to the circuit.
- Jumper wires (M-F, M-M) – Make connections between components.
- Breadboard – A board for building circuits without soldering.
- Lighter (Simulating Gas) – Used to release gas.
- Container (Simulating Cylinder) – Represents an LPG cylinder.
- Soldering Iron – Join electronic components permanently.
- Screwdriver Set – Fix hardware components.

Software Tools: Arduino IDE

Cost Breakdown:

Component Name	Quantity	Price (BDT)
Arduino Uno R3	1	949
Load Cell Sensor (10KG)	1	380
HX711 ADC Module	1	150
MQ-2 Gas Sensor	1	290
I2C LCD 20x4	1	540
5V Cooling Fan	1	119
5V Relay Module	1	75
Active Buzzer 5V	1	15
Adapter DC 5V 2A	1	300
Misc (Wires, Breadboard)	1 Set	288

Communication Protocol:

1. I2C (Inter-Integrated Circuit)

I2C is extensively used in the project to establish communication between the main microcontroller (Arduino Uno R3) and its sub-modules.

LCD: The 20x4 LCD is connected to an I2C bus for receiving information to display. It is attached to the A4 (SDA) and A5 (SCL) pins of the Arduino.

2. Two-Wire Serial (HX711)

The weight monitoring hardware is connected using a particular type of serial communication:

Load Cell & HX711: The HX711 amplifier, that amplifies the output from the load cell, is connected to the Arduino by a two wire serial interface, a Data line (DT) and a Clock line (SCK).

Pin Mapping: Here, the Data line (DT) is connected to digital pin D2, and the Clock line (SCK) to digital pin D3.

3. Analog Signal Transmission

Although not a true "digital protocol" in terms of computer networking, sensor data is transmitted using analog signals:

MQ-2 Gas Sensor: This sensor provides an analog signal (voltage) corresponding to the concentration of gas present. This is connected to the Arduino's A0 analog pin, which converts it into a digital signal.

4. Digital Output (GPIO Control)

The system also employs basic digital (HIGH/LOW) signals to activate safety devices:

Buzzer: Controlled by a digital output (D8).

Relay/Exhaust Fan: Triggered via digital output on pin D7 to turn fan on/off in case of a leak.

Implementation:

Step 1: System Startup The controller is booted up, powering up the display and other hardware, and tares (zeros) the load cell in order to calibrate the initial weight.

Step 2: Data Collection: The MQ-2 sensor samples the live gas level (combustible gas concentration) of the microcontroller and the HX711 samples the real-time weight of the cylinder.

Step 3: Live Display: LCD display shows live weight and raw gas level data.

Step 4: Primary Safety Check (Gas Leak): If the gas reading is above the safety threshold, the system will trigger the emergency alarm and the fan (via the relay) instantly and display an evacuation warning message which will continue to loop until the environment is clear.

Step 5: Secondary Check (Low Fuel) If safe, the system will check the weight. If below the threshold (for example 0.30 kg) it is a visual warning of "Low Gas!" on the LCD but no audio alarm.

Step 6: Repeat The system sleeps for a short period of time to avoid LCD flicker, then immediately repeats. This continues indefinitely.

Challenges:

1. Load Cell Calibration

Challenge: The load cell sensor was hard to calibrate. So it was hard to obtain accurate measurements. The code we used had some issues measuring weight, so the weight output was inaccurate.

How we solved the problem: We updated the calibration part of the code. We put various known weights on the scale and increased the variable called the calibration factor in the code until the weight shown was accurate.

2. HX711 Integrity

Challenge: The system sometimes glitched because of loose pins on the HX711 amplifier.

How we solved it: We used a soldering iron to firmly attach the header pins and load cell wires directly to the HX711 board.

4. Sustainability & Impact

Sustainability

Energy Efficiency:
The system is powered by a 5V, 2A power source, with a maximum power consumption of about 10 watts. The Arduino Uno consumes less than 50mA when operating, and the MQ-2 sensor is in a low power mode between measurements. In addition, the exhaust fan and buzzer are only used when necessary; it basically means that the system is on low power mode most of the time and this allows the system to be left on 24 hours a day without any extra use of power, even in households where there are low supplies.

Recyclable and Reusable Materials:
The breadboard and jumper wires we used are reusable and can be used for other projects if needed. The Arduino Uno is a reusable development system that can be used for other projects as well. For the load cell, HX711 and LCD have a long lifespan and the MQ-2 sensor is a consumable part of the system due to the heated element but it may last for a few years. The majority of system components are readily recyclable via electronic waste programs.

Impact

Societal Impact:
In Bangladesh and almost all of South Asia, most of the people are injured and houses are damaged by LPG accidents each year. And for that an affordable detection system like this will help to minimize the damage, particularly in households who can't afford high end smart detectors.

Industrial:
In industrial environments such as restaurants, food processing facilities, or laboratories that use LPG or other combustible gases, a larger scale version of the system can be part of a comprehensive safety system, as the system is built on the open source Arduino platform, its design can be readily shared and modified by anyone having proper knowledge.

Future Work

● Wi-Fi and IoT Connectivity: By integrating an ESP8266 or ESP32 module it will enable the system to send realtime notifications to a mobile phone using an app and it'll even allow remote monitoring even though someone is not in their houses.
● Gas Detection: Using a more sensitive LPG sensor like the MQ-6 instead of the MQ-2 would make it better as the system's ability to detect LPG avoids any false alarms from other gas sources such as smoke and any unknown fumes if it happens.

Limitations

Hardware Limitations:

● Load Cell Sensitivity: While we use TAL220 10kg load cell it can be slightly jittery if the cylinder is not placed correctly and it might cause issues with the measurements.
● MQ-2 Sensitivity: MQ-2 is not only sensitive to LPG it also responds to smoke, alcohol vapor, or other flammable gases, which could lead to false alarms in the kitchen due to fumes.
● Lack of Real Time Clock: The system is not equipped with a clock so it can’t record the time of the leak and how long it lasted.

Software Limitations:

● Hardcoded Thresholds: The threshold for a low gas warning and the threshold for a gas leakage warning are programmed into the Arduino code but a different size gas tank or different requirements would mean reprogramming the board, which is not available to the user who are non technical.

● Sequential Processing: As Arduino Uno has a single loop so the buzzer alarm loop may introduce a slight delay which may delay response to other events. Therefore, a real time operating system (RTOS) or interrupts would make it more responsive.

5. Results & Discussion

The system has been tested in various scenarios to ensure that the sensors are operating properly and the software logic is correct. The testing was divided into several stages such as testing Emergency system in the case of a gas leak, and testing to see if the weight of the tank is too low.

Gas Sensor (MQ-2) Parameters:

Trigger Threshold- 600

Simulated Leak Source- Butane lighter

Load Cell Parameters:

Trigger Threshold- 0.30 kg

Reference Weight- 0.52 kg (Water bottle)

MQ-2 Gas Sensor Testing:

Test Condition	Sensor Reading	Threshold	Buzzer State	Relay/Fan State	LCD Status Output
Clean Ambient Air	330	600	OFF	OFF	"Status: Safe"
Simulated Leak	850+	600	ON	ON	"!! GAS LEAK !!"

Load Cell Sensor Testing:

Actual Object Weight	LCD Displayed Weight	Low Gas Threshold	Visual Warning State
Empty Scale	0.00 kg	0.30 kg	Inactive
Bottle	0.52 kg	0.30 kg	Inactive
Bottle	0.25kg	0.30 kg	Active ("Low Gas!")

Through testing, it appears that the system runs efficiently, which means there are fewer risks and more accurate level monitoring. For example, when examining for leaks, the MQ-2 sensors triggered when the gas level was over 600. However, when it got below 600 again, the fan and the buzzer stopped. The weight scale indicated there was a total of 0.50+- kg so this indicates that internal logic has worked effectively with respect to the fuel warning system.

6. Conclusion

Project Recap and Main Findings:

Our LPG Safety system worked. It shut down gas leaks and measured the fuel consumption at the same time, using a microcontroller. The LPG Safety System was able to do this because it contained gas sensors and a weighing scale.

At first we had some issues. The hardware connections were not secure. We solved this by soldering the connections. We also had problems with the load cell calibration but we were able to resolve that. The system is able to detect leaks and turn on the fan and alarm. At the same time it can measure the amount of gas and indicate when it is running low.

Sustainability and Broader Impact:

Apart from its success this system aids in resource conservation and safety. This is done by providing gas level information. This helps in not wasting LPG cylinders. It also aids in refilling the gas cylinders. The automatic ventilation system in emergency situations minimises the fire, explosion and pollution hazards. This approach to disaster management helps to build resilient and smart infrastructure.

7. Contribution

Name	ID	Role(s) / Responsibilities	Specific Contributions
Plaban Basu	22101831	System Architecture & Design	• Designed the overall system and hardware connection maps. • Created the workflow flowchart.
Abdullah Al Mahamud	22101811	Hardware Assembly	• Assembled the physical circuit, including wiring the components. • Calibrated components.
Bishan Paul	21201714	Testing & Documentation	• Tested system under normal and gas leakage conditions. • Verified sensor accuracy of MQ-2 and load cell. • Wrote Results, Limitations and Sustainability.
Torsha Bashar Chamak	21301083	Procurement and Assembly	• Component procurement. • Hardware assembly.
Tamjeed Abdullah Noor	19201062	Embedded Software Development	• Handled the software part of Arduino. • Integrated the HX711, LCD and MQ-2 libraries.

References

Oo, Z. L., Lai, T. W., & Moe, A. (2021). *IoT based LPG gas level detection & gas leakage accident prevention with alert system*. ResearchGate.